Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6541057198

FURTHER MATHEMATICS **9231/33**

Paper 3 Further Mechanics **October/November 2023**

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use $10\,\text{m}\,\text{s}^{-2}$.

INFORMATION
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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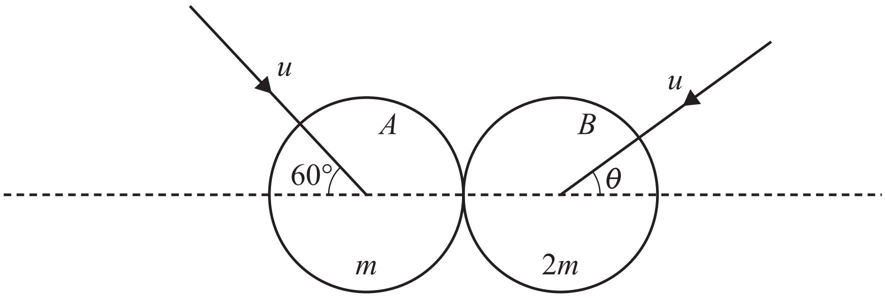
1



Two uniform smooth spheres A and B of equal radii have masses m and $2m$ respectively. The two spheres are moving with equal speeds u on a smooth horizontal surface when they collide. Immediately before the collision, A's direction of motion makes an angle of 60° with the line of centres, and B's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is e.

After the collision, the component of the velocity of A along the line of centres is v and B moves perpendicular to the line of centres. Sphere A now has twice as much kinetic energy as sphere B.

(a) Show that $v = \frac{1}{2}u(4\cos\theta - 1)$. [1]

By CoM: (horizontally) A must change change direction

$mu\cos 60 - 2mu\cos\theta = m(-v)$ ⇒ where rightward is positive

$v = 2u\cos\theta - \frac{u}{2}$

$\boxed{v = \frac{1}{2}u\left(4\cos\theta - 1\right)}$

(b) Find the value of $\cos\theta$. [4]

sphere A KE:

$= \frac{1}{2}m\left(\sqrt{v^2 + (u\sin 60)^2}\right)^2$

$= \frac{1}{2}m\left(v^2 + u^2\left(\frac{3}{4}\right)\right)$

$= \frac{1}{2}m\left(\left(\frac{u}{2}\left(4\cos\theta - 1\right)\right)^2 + \frac{3u^2}{4}\right)$

$\frac{u^2}{4}\left(16\cos^2\theta - 8\cos\theta + 1\right)$

KE_B:

$\frac{1}{2}(2m)\left(u^2\sin^2\theta\right)$

$= mu^2\sin^2\theta$

$2\cancel{m}v^2\sin^2\theta = \frac{1}{2}\cancel{m}\left(\left(2v\cos\theta - \frac{v}{2}\right)^2 + \frac{3v^2}{4}\right)$

$4\cancel{v^2}\sin^2\theta = 4\cancel{v^2}\cos^2\theta + \frac{v^2}{4} - 2v^2\cos\theta + \frac{3v^2}{4}$

$4(1-\cos^2\theta) = 4\cos^2\theta - 2\cos\theta + 1$

$0 = 8\cos^2\theta - 2\cos\theta - 3$

$= (4\cos\theta - 3)(2\cos\theta + 1)$

$\cos\theta = 3/4$ or $-1/2$

We know $v_{(left)} > 0$

$\frac{1}{2}(4v\cos\theta - 1) > 0$

only when

$\boxed{\cos\theta = 3/4}$

(c) Find the value of e. [2]

$e = \dfrac{v - 0}{v\cos 60 + v\cos\theta} = \dfrac{\frac{\cancel{v}}{2}\left(4\left(\frac{3}{4}\right) - 1\right)}{\frac{\cancel{v}}{2} + \frac{3\cancel{v}}{4}} = \boxed{\frac{4}{5}}$

2 A ball of mass 2 kg is projected vertically downwards with speed $5\,\mathrm{m\,s^{-1}}$ through a liquid. At time t s after projection, the velocity of the ball is $v\,\mathrm{m\,s^{-1}}$ and its displacement from its starting point is x m. The forces acting on the ball are its weight and a resistive force of magnitude $0.2v^2$ N.

(a) Find an expression for v in terms of t. [6]

$m\frac{dv}{dt} = mg - 0.2v^2$

$\frac{dv}{dt} = g - 0.1v^2$

$\frac{dv}{100 - v^2} = \frac{1}{10}dt$

$\int \frac{1}{(10-v)(10+v)} dv = \int \frac{1}{10} dt$

$\frac{1}{(10-v)(10+v)} = \frac{A}{(10-v)} + \frac{B}{(10+v)}$

$\hookrightarrow 10A + 10vA + 10B - 10vB$

$\hookrightarrow 10A - 10B = 0 \Rightarrow A = B$

$10A + 10B = 1 \Rightarrow A = B = \frac{1}{20}$

$\Rightarrow \frac{1}{20}\left(\int \frac{1}{10-v} dv + \int \frac{1}{10+v} dv\right) = \int \frac{1}{10} dt$

$$\frac{\ln(10-v)}{(-1)} + \frac{\ln(10+v)}{(1)} = 2t + c$$

$$\ln\left(\frac{10+v}{10-v}\right) = 2t+c$$

↳ subs $(t, v) = (0, 5)$

$\ln 3 = c$

$$\frac{10+v}{10-v} = 3e^{2t}$$

$10+v = 30e^{2t} - 3ve^{2t}$

$v(1+3e^{2t}) = 30e^{2t} - 10$

$$\boxed{v = \frac{10(3e^{2t}-1)}{3e^{2t}+1}}$$

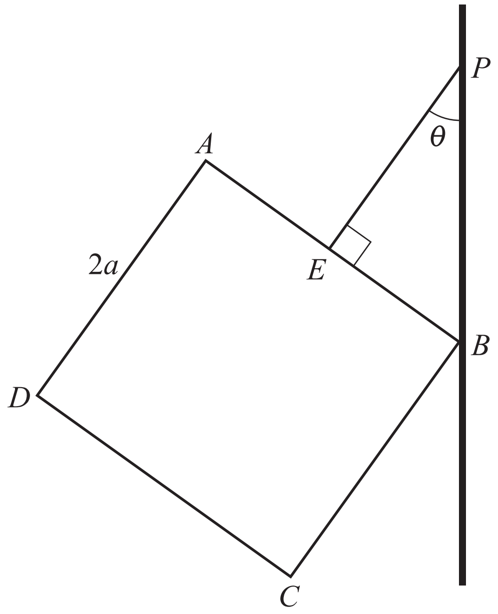
(b) Deduce what happens to v for large values of t. [1]

$$v = \frac{10(3 - e^{-2t})}{3 + e^{-2t}}$$

↳ as $t \to \infty$

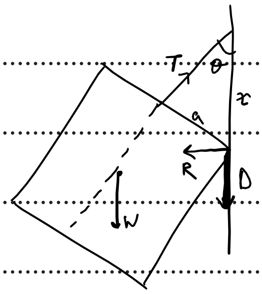
$$= \frac{10(3)}{3} \quad \boxed{= 10\,\text{ms}^{-1}}$$

3



A uniform square lamina of side $2a$ and weight W is suspended from a light inextensible string attached to the midpoint E of the side AB. The other end of the string is attached to a fixed point P on a rough vertical wall. The vertex B of the lamina is in contact with the wall. The string EP is perpendicular to the side AB and makes an angle θ with the wall (see diagram). The string and the lamina are in a vertical plane perpendicular to the wall. The coefficient of friction between the wall and the lamina is $\frac{1}{2}$.

Given that the vertex B is about to slip up the wall, find the value of $\tan\theta$. [8]



Translationally:

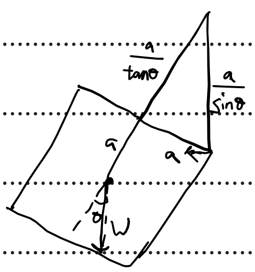
$T\cos\theta = W + D \Rightarrow T\cos\theta = W + \frac{R}{2}$ (1)

$T\sin\theta = R$ (2)

$D = \mu R$

$D = \frac{R}{2}$

Rotationally:



$W\sin\theta\left(\frac{a}{\tan\theta} + a\right) = \frac{Ra}{\sin\theta}$ (3)

Subs ② in ③

$$W\sin\theta\left(\frac{a}{\tan\theta}+a\right) = Ta$$

$$W\sin\theta\left(\frac{\cos\theta}{\sin\theta}+1\right) = T$$

With ① & ②:

$$T\cos\theta - \frac{T\sin\theta}{2} = W$$

$$\cancel{T}\left(\cos\theta - \frac{\sin\theta}{2}\right)\left(\cos\theta + \sin\theta\right) = \cancel{T}$$

$$\cos^2\theta + \frac{\sin\theta\cos\theta}{2} - \frac{\sin^2\theta}{2} = 1$$

$$1 - \sin^2\theta + \frac{\sin\theta\cos\theta}{2} - \frac{\sin^2\theta}{2} = 1$$

$$\frac{-3\sin^2\theta}{2} + \frac{\sin\theta\cos\theta}{2} = 0$$

$$3\sin^2\theta - \sin\theta\cos\theta = 0$$

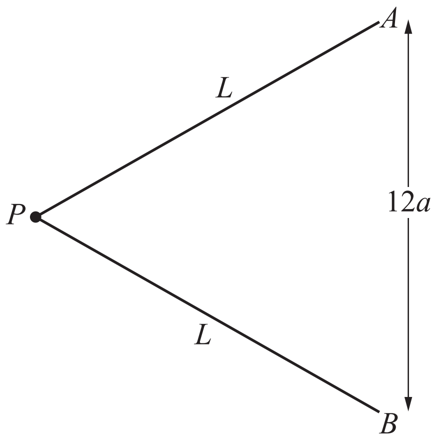
$$\sin\theta\,(3\sin\theta - \cos\theta) = 0$$

$\hookrightarrow \sin\theta = 0 \Rightarrow$ evidently, impossible

$\hookrightarrow 3\sin\theta = \cos\theta$

$\hookrightarrow \boxed{\tan\theta = 1/3}$

4



A light elastic string has natural length $8a$ and modulus of elasticity $5mg$. A particle P of mass m is attached to the midpoint of the string. The ends of the string are attached to points A and B which are a distance $12a$ apart on a smooth horizontal table. The particle P is held on the table so that $AP = BP = L$ (see diagram). The particle P is released from rest. When P is at the midpoint of AB it has speed $\sqrt{80ag}$.

(a) Find L in terms of a. [5]

Initially:

$EPE = \frac{\lambda x^2}{2l}$

$= \frac{(5mg)(2L-8a)^2}{16a}$

$KE = 0$

Finally:

$EPE = \frac{\lambda x^2}{2l}$

$= \frac{5mg(4a)^2}{2(8a)}$

$= 5mg\,a$

$KE = \frac{1}{2}m\left(\sqrt{80ga}\right)^2 = 40mga$

By CoE:

$$\left(\frac{5mg}{16a}\right)(2L-8a)^2 = 40mga + 5mga$$

$$(2L-8a)^2 = 16a(9a) = 144a^2$$

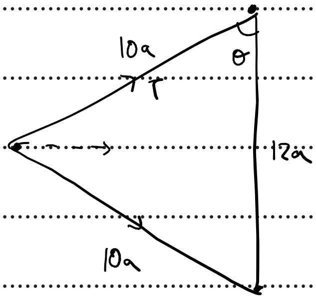
$2L - 8a = \pm 12a \Rightarrow$ not $= -12a$

↳ then $L<0$

↳ $2L - 8a = 12a$

$$\boxed{L = 10a}$$

(b) Find the initial acceleration of P in terms of g. [3]



$F = 2T\sin\theta$

↳ $\sin\theta = 0.8$

↳ $T = \frac{\lambda x}{l} = \frac{(5mg)(\cancel{12a})^{3}}{{}_{2}\cancel{8a}}$

$= 7.5mg$

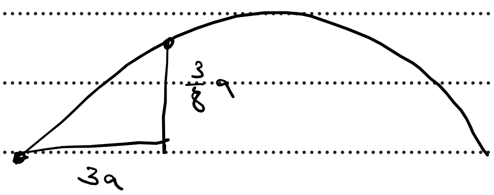
$F = 2(7.5mg)(0.8)$

$= 12mg$

↳ $\boxed{a = 12g}$

5 A particle P is projected with speed $u\,\mathrm{m\,s^{-1}}$ at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. During its flight P passes through the point which is a horizontal distance $3a$ from O and a vertical distance $\frac{3}{8}a$ above the horizontal plane. It is given that $\tan\theta = \frac{1}{3}$.

(a) Show that $u^2 = 8ag$. [2]



$y = x\tan\theta - \frac{gx^2}{2u^2}\sec^2\theta$

↳ sub $(x, y, \tan\theta) = (3a, \frac{3}{8}a, \frac{1}{3})$

$\frac{3}{8}a = 3a\left(\frac{1}{3}\right) - \frac{g(9a^2)}{2u^2}(\tan^2\theta + 1)$

$\frac{9ga^2}{8u^2}\left(\frac{10}{9}\right) = \frac{5}{8}a$

$5ga^2 = \frac{5}{8}u^2a$

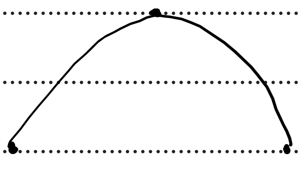
↳ $\boxed{u^2 = 8ga}$

A particle Q is projected with speed $V\,\mathrm{m\,s^{-1}}$ at an angle α above the horizontal from O at the instant when P is at its highest point. Particles P and Q both land at the same point on the horizontal plane at the same time.

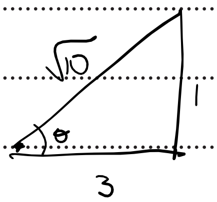
(b) Find V in terms of a and g. [7]

for P:

$\tan\theta = \frac{1}{3}$, $u = \sqrt{8ga}$



We know that horizontally, Q is twice as fast as P



↳ $V\cos\alpha = 2\sqrt{8ga}\cos\left(\tan^{-1}\left(\frac{1}{3}\right)\right)$

$V\cos\alpha = \sqrt{28.8ga}$ (1)

@ time $=t$, height of $P=0$

↳ $s = vt + \frac{1}{2}at^2$

$0 = (\sqrt{8ga}\sin\theta)(t) - \frac{1}{2}g(t)^2$

$\sqrt{8ga}\left(\frac{1}{\sqrt{10}}\right) = 5t$

$t = \frac{\sqrt{0.8ga}}{5}$

For Q:

↳ it reaches $s=0$ in half the time

$0 = v\left(\frac{t}{2}\right)^2 + \frac{1}{2}(a)\left(\frac{t}{2}\right)^2$

$= V\sin\alpha\left(\cancel{\frac{\sqrt{0.8ga}}{10}}\right) - \frac{1}{2}g\left(\frac{\sqrt{0.8ga}}{10}\right)^2$

$V\sin\alpha = 5\left(\frac{\sqrt{0.8ga}}{10}\right)$

$= \sqrt{0.2ga}$

$V = \sqrt{(V\sin\alpha)^2 + (V\cos\alpha)^2}$

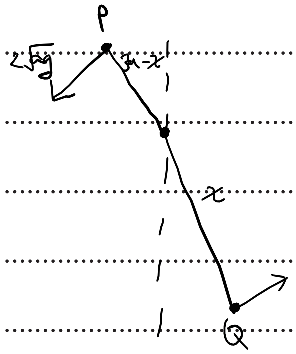
$= \sqrt{28.8ga + 0.2ga}$

$\boxed{= \sqrt{29ga}}$

6 A particle P of mass m is attached to one end of a light inextensible rod of length $3a$. An identical particle Q is attached to the other end of the rod. The rod is smoothly pivoted at a point O on the rod, where $OQ = x$. The system, of rod and particles, rotates about O in a vertical plane.

At an instant when the rod is vertical, with P above Q, the particle P is moving horizontally with speed u. When the rod has turned through an angle of 60° from the vertical, the speed of P is $2\sqrt{ag}$, and the tensions in the two parts of the rod, OP and OQ, have equal magnitudes.

(a) Show that the speed of Q when the rod has turned through an angle of 60° from the vertical is $\frac{2x}{3a-x}\sqrt{ag}$. [2]



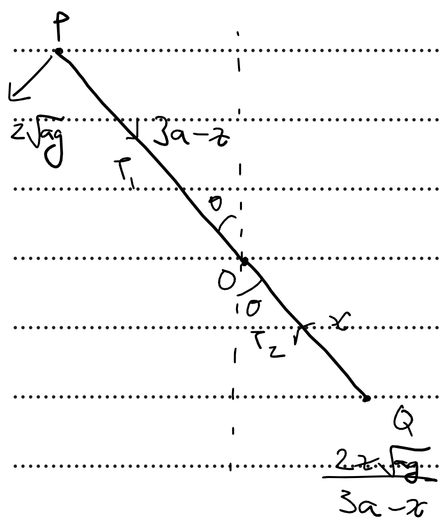
ω is constant for the system

$$\frac{v_p}{r_p} = \frac{v_q}{r_q}$$

$$\frac{2\sqrt{ag}}{3a-x} = \frac{v_q}{x}$$

$$\hookrightarrow v_q = \frac{2x\sqrt{ag}}{3a-x}$$

(b) Find x in terms of a. [5]



@ P:

$$mg\cos\theta + T_1 = \frac{mv^2}{r}$$

$$T_1 = \frac{m(4ag)}{3a-x} - mg\cos 60$$

$$= \frac{4mga}{3a-x} - \frac{mg}{2}$$

@ Q:

$$\frac{mv^2}{r} = T_2 - mg\cos\theta$$

$$T_2 = \frac{m(4x^2ag)}{(3a-x)^2 x} + \frac{mg}{2}$$

$T_1 = T_2$

$\hookrightarrow \dfrac{4mga}{3a-x} - \dfrac{mg}{2} = \dfrac{4xmga}{(3a-x)^2} + \dfrac{mg}{2}$

$\dfrac{(3a-x)4a - 4xa}{(3a-x)^2} = 1$

$12a^2 - 4ax - 4xa = (3a-x)^2$

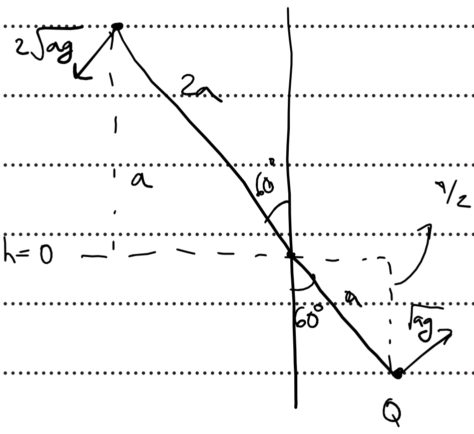
$12a^2 - 8ax = 9a^2 + x^2 - 6ax$

$3a^2 - 2ax - x^2 = 0$

$(a - x)(3a + x)$

$\hookrightarrow \boxed{x = a}$

(c) Find u in terms of a and g. [4]



At 60°:

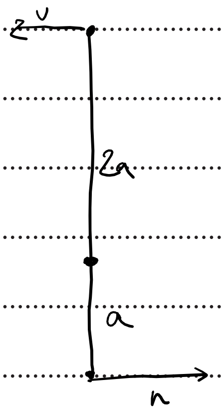
$PE = mga + mg\left(-\frac{a}{2}\right) = \frac{mga}{2}$

$KE = \frac{1}{2}m(2\sqrt{ag})^2 + \frac{1}{2}m(\sqrt{ag})^2$

$= 2mga + \frac{1}{2}mga$

$= \frac{5}{2}mga$

$\hookrightarrow$ Total energy $= 3mga$



@ $\theta = 0°$

$PE = -mga + 2mga$

$= mga$

$KE = 3mga - mga = 2mga$

$\rightarrow v = 2n$

$\hookrightarrow 2mga = \frac{1}{2}m\left(v^2 + (n)^2\right)$

$2mga = \frac{1}{2}m\left(\frac{5v^2}{4}\right)$

$v^2 = \frac{16}{5}ga \Rightarrow \boxed{v = \frac{4}{\sqrt{5}}ga \ \ ms^{-1}}$

Additional page

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